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Vehicle seating assembly with lockout cam

Abstract

A vehicle seating assembly includes a mounting bracket, a seat, a seatback, and a cam assembly. The seat is rotatably coupled to the mounting bracket. The seatback is rotatably coupled to the mounting bracket. The seatback includes a protrusion. The cam assembly is rotatably coupled to the mounting bracket. The cam assembly includes a lockout cam having a first end and a second end. The first end of the lockout cam engages with the protrusion of the seatback such that movement of the seatback relative to the mounting bracket is restricted in a first non-use state. The second end of the lockout cam engages with the seat such that movement of the seat relative to the mounting bracket is restricted in a second non-use state.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to vehicles. More specifically, the present disclosure relates to vehicle seating assemblies.

BACKGROUND OF THE DISCLOSURE

(2) Consumers often compare available features and functionality between vehicles when making a purchasing decision. Accordingly, additional solutions are needed that provide features and functionality that are desirable to consumers.

SUMMARY OF THE DISCLOSURE

(3) According to a first aspect of the present disclosure, a vehicle seating assembly includes a mounting bracket, a seat, a seatback, and a cam assembly. The seat is rotatably coupled to the mounting bracket. The seatback is rotatably coupled to the mounting bracket. The seatback includes a protrusion. The cam assembly is rotatably coupled to the mounting bracket. The cam assembly includes a lockout cam having a first end and a second end. The first end of the lockout cam engages with the protrusion of the seatback such that movement of the seatback relative to the mounting bracket is restricted in a first non-use state. The second end of the lockout cam engages with the seat such that movement of the seat relative to the mounting bracket is restricted in a second non-use state.

(4) Embodiments of the first aspect of the disclosure can include any one or a combination of the following features: the lockout cam is biased toward an engaged position; the engaged position of the lockout cam is defined as physical contact between the first end of the lockout cam and the protrusion of the seatback in the first non-use state; the engaged position of the lockout cam is defined as physical contact between the second end of the lockout cam and the seat in the second

non-use state; an actuation cam that is configured to engage with a stationary structure below the vehicle seating assembly, wherein a position of the vehicle seating assembly relative to the stationary structure dictates a position of the actuation cam; the stationary structure is at least a portion of a track assembly to which the vehicle seating assembly is configured to be mounted; a cable that extends between the actuation cam and the lockout cam such that the position of the actuation cam influences whether the lockout cam is in the engaged position or a disengaged position; the actuation cam is configured to disengage with a top surface of the stationary structure when the vehicle seating assembly is in at least one of the first non-use state and the second non-use state; engagement between the actuation cam and a top surface of the stationary structure places the actuation cam in a position that provides a force to the cable that counter-acts the biasing of the lockout cam to the engaged position; disengagement of the actuation cam with the top surface of the stationary structure causes a slackening of the cable that extends between the actuation cam and the lockout cam, and wherein such slackening of the cable allows the biasing of the lockout cam to place the lockout cam in the engaged position; the first non-use state results in actuation of a track release button and an interlock release button of a track assembly to which the vehicle seating assembly is configured to be mounted; the second non-use state results in actuation of the interlock release button; the first non-use state places the seatback in a position that prevents occupancy of the vehicle seating assembly; and the second non-use state places the seat in a position that prevents occupancy of the vehicle seating assembly.

(5) According to a second aspect of the present disclosure, a vehicle seating assembly includes a mounting bracket, a seat, a seatback, a cam assembly, an actuation cam, and a cable. The seat is rotatably coupled to the mounting bracket. The seatback is rotatably coupled to the mounting bracket. The seatback includes a protrusion. The cam assembly is rotatably coupled to the mounting bracket. The cam assembly includes a lockout cam that is biased toward an engaged position. The lockout cam includes a first end and a second end. The first end of the lockout cam engages with the protrusion of the seatback such that movement of the seatback relative to the mounting bracket is restricted in a first non-use state. The engaged position of the lockout cam is defined as physical contact between the first end of the lockout cam and the protrusion of the seatback in the first non-use state. The second end of the lockout cam engages with the seat such that movement of the seat relative to the mounting bracket is restricted in a second non-use state. The engaged position of the lockout cam is defined as physical contact between the second end of the lockout cam and the seat in the second non-use state. The actuation cam engages with a stationary structure below the vehicle seating assembly. A position of the vehicle seating assembly relative to the stationary structure dictates a position of the actuation cam. The actuation cam disengages with a top surface of the stationary structure when the vehicle seating assembly is in at least one of the first non-use state and the second non-use state. Engagement between the actuation cam and the top surface of the stationary structure places the actuation cam in a position that provides a force to the cable that counter-acts the biasing of the lockout cam to the engaged position. The cable extends between the actuation cam and the lockout cam such that the position of the actuation cam influences whether the lockout cam is in the engaged position or a disengaged position.

(6) Embodiments of the second aspect of the present disclosure can include any one or a combination of the following features: the stationary structure is at least a portion of a track assembly to which the vehicle seating assembly is mounted; disengagement of the actuation cam with the top surface of the stationary structure causes a slackening of the cable that extends between the actuation cam and the lockout cam, and wherein such slackening of the cable allows the biasing of the lockout cam to place the lockout cam in the engaged position; the first non-use state results in actuation of a track release button and an interlock release button of a track assembly to which the vehicle seating assembly is mounted; the second non-use state results in actuation of the interlock release button; and the first non-use state places the seatback in a position

that prevents occupancy of the vehicle seating assembly, whereas the second non-use state places the seat in a position that prevents occupancy of the vehicle seating assembly.

(7) These and other aspects, objects, and features of the present disclosure will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 is a front perspective view of a passenger compartment of a vehicle, illustrating vehicle seating assemblies positioned therein, according to one example;

(3) FIG. 2A is a schematic representation of movement of the vehicle seating assembly within an occupancy region of the passenger compartment, according to one example;

(4) FIG. 2B is a schematic representation of movement of the vehicle seating assembly outside of the occupancy region of the passenger compartment, according to one example;

(5) FIG. 2C is a schematic representation of movement of the vehicle seating assembly outside of the occupancy region of the passenger compartment, according to another example;

(6) FIG. 3 is a front perspective view of the vehicle seating assembly, according to one example;

(7) FIG. 4 is an expanded view of the vehicle seating assembly, from region IV of FIG. 3, according to one example;

(8) FIG. 5 is a side view of a lower portion of the vehicle seating assembly, illustrating interaction between components of the vehicle seating assembly and a track assembly, according to one example;

(9) FIG. 6 is a cross-sectional view of the vehicle seating assembly and the track assembly, taken at line VI-VI of FIG. 5, illustrating interaction between components of the vehicle seating assembly, according to one example;

(10) FIG. 7 is an expanded rear perspective view of region VII of FIG. 3, illustrating a lockout cam of the vehicle seating assembly, according to one example;

(11) FIG. 8A is rear perspective view of the lower region of the vehicle seating assembly, illustrating an easy entry release cam and an interlock release cam, according to one example;

(12) FIG. 8B is a front perspective view of the lower region of the vehicle seating assembly, illustrating an actuation arm and an actuation lever, according to one example;

(13) FIG. 9A is a rear perspective view of the lower region of the vehicle seating assembly, illustrating actuation of a track release button by the actuation lever, according to one example;

(14) FIG. 9B is a front perspective view of the lower region of the vehicle seating assembly, illustrating actuation of the track release button by the actuation lever, according to one example;

(15) FIG. 10 is a rear perspective view of the vehicle seating assembly, illustrating a seat of the vehicle seating assembly in a raised position and the lockout cam in an engaged position with the seat, according to one example;

(16) FIG. 11A is a rear perspective view of the lower region of the vehicle seating assembly, illustrating actuation of an interlock release button by the actuation arm, according to one example;

(17) FIG. 11B is a front perspective view of the lower region of the vehicle seating assembly, illustrating actuation of the interlock release button by the actuation arm, according to one example;

(18) FIG. 12 is a side view of the lower region of the vehicle seating assembly, illustrating an actuation cam in a disengaged position relative to a stationary structure that is positioned below the vehicle seating assembly, according to one example;

(19) FIG. 13 is a rear perspective view of the vehicle seating assembly, illustrating a seatback of the

vehicle seating assembly in a forwardly-rotated position and the lockout cam in an engaged position with the seatback, according to one example;

(20) FIG. 14A is a rear perspective view of the lower region of the vehicle seating assembly, illustrating actuation of the track release button and the interlock release button, according to one example;

(21) FIG. 14B is a front perspective view of the lower region of the vehicle seating assembly, illustrating actuation of the track release button and the interlock release button, according to one example;

(22) FIG. 15A is side view of the lower region of the vehicle seating assembly, illustrating the actuation cam in an engaged position with the stationary structure that is positioned below the vehicle seating assembly, according to one example; and

(23) FIG. 15B is a rear perspective view of the vehicle seating assembly, illustrating the lockout cam in a disengaged position, according to one example.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(24) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof shall relate to the concepts as oriented in FIG. 1. However, it is to be understood that the concepts may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(25) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to a vehicle seating assembly. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.

(26) As used herein, the term “and/or,” when used in a list of two or more items, means that any one of the listed items can be employed by itself, or any combination of two or more of the listed items, can be employed. For example, if a composition is described as containing components A, B, and/or C, the composition can contain A alone; B alone; C alone; A and B in combination; A and C in combination; B and C in combination; or A, B, and C in combination.

(27) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(28) As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art. When the term “about” is used in describing a value or an end-point of a range, the disclosure should be understood to include the specific value or end-point referred to. Whether or not a numerical value or end-point of a range in the specification recites “about,” the numerical value or end-point of a range is

intended to include two embodiments: one modified by “about,” and one not modified by “about.” It will be further understood that the end-points of each of the ranges are significant both in relation to the other end-point, and independently of the other end-point.

(29) The terms “substantial,” “substantially,” and variations thereof as used herein are intended to note that a described feature is equal or approximately equal to a value or description. For example, a “substantially planar” surface is intended to denote a surface that is planar or approximately planar. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. In some embodiments, “substantially” may denote values within about 10% of each other, such as within about 5% of each other, or within about 2% of each other.

(30) As used herein the terms “the,” “a,” or “an,” mean “at least one,” and should not be limited to “only one” unless explicitly indicated to the contrary. Thus, for example, reference to “a component” includes embodiments having two or more such components unless the context clearly indicates otherwise.

(31) Referring to FIGS. 1-15B, reference numeral **20** generally designates a vehicle **20**. The vehicle **20** includes a passenger compartment **24** that is provided with one or more vehicle seating assemblies **28**. The vehicle seating assembly **28** includes a mounting bracket **32**, a seat **36**, a seatback **40**, and a cam assembly **44**. The seat **36** is rotatably coupled to the mounting bracket **32**. The seatback **40** is rotatably coupled to the mounting bracket **32**. The seatback **40** includes a protrusion **48**. The cam assembly **44** is rotatably coupled to the mounting bracket **32**. The cam assembly **44** includes a lockout cam **52** having a first end **56** and a second end **60**. The first end **56** of the lockout cam **52** engages with the protrusion **48** of the seatback **40** such that movement of the seatback **40** relative to the mounting bracket **32** is restricted in a first non-use state (e.g., see FIG. 13). The second end **60** of the lockout cam **52** engages with the seat **36** such that movement of the seat **36** relative to the mounting bracket **32** is restricted in a second non-use state (e.g., see FIG. 10).

(32) Referring again to FIGS. 1-2C, the vehicle seating assembly **28** is coupled to a floor **64** of the passenger compartment **24**. For example, one or more track assemblies **68** may couple the vehicle seating assembly **28** to the floor **64** of the passenger compartment **24**. In such examples, the track assembly **68** can include a lower track **72** that is directly coupled to the floor **64** and an upper track **76** that is indirectly coupled to the floor **64**. The upper track **76** is received within a slot **80** defined by the lower track **72**. The upper track **76** is capable of moving along the lower track **72** such that a longitudinal position of the vehicle seating assembly **28** within the passenger compartment **24** may be adjusted, as indicated by arrow **84**. The upper track **76** may alternatively be referred to as a carriage assembly. The vehicle seating assembly **28** is coupled to the upper track **76**. For example, anchors **88** may extend from the vehicle seating assembly **28** for coupling to the upper track **76**. More specifically, one of the anchors **88** may extend from the mounting bracket **32** on each side of the vehicle seating assembly **28** to engage with a corresponding one of the upper tracks **76**. Therefore, movement of the upper track **76** and the vehicle seating assembly **28** may be coupled such that movement of one of the upper track **76** or the vehicle seating assembly **28** in the longitudinal direction along the lower track **72** results in corresponding movement of the other of the upper track **76** or the vehicle seating assembly **28**.

(33) Referring further to FIGS. 1-2C, the vehicle seating assembly **28** of the present disclosure is configured to provide an end user with a substantial degree of adjustability. More specifically, as depicted in FIG. 2A, the vehicle seating assembly **28** is capable of adjusting the longitudinal position along the lower track **72** within an occupancy region **92** to enable the user to adjust a positioning of the vehicle seating assembly **28** within the passenger compartment **24** according to their specific comfort preferences. Additionally, the vehicle seating assembly **28** is capable of adjusting the longitudinal position along the lower track **72** beyond the occupancy region **92**. For example, the vehicle seating assembly **28** is capable of assuming a storage configuration, as depicted in FIG. 2B, where the seat **36** is rotated upward toward the seatback **40** to make available

a greater surface area of the floor **64** of the passenger compartment **24**. Further, the vehicle seating assembly **28** is capable of assuming an easy-entry configuration by adjusting the longitudinal position along the lower track **72** and rotating the seatback **40** toward the seat **36**, as shown in FIG. 2C.

(34) With specific reference to FIGS. 2A-2C, the vehicle seating assembly **28** is intended to be occupied when in a use state, as depicted in FIG. 2A. In the use state, the vehicle seating assembly **28** is arranged to receive an occupant on the seat **36** thereof. The use state is not a single position of the vehicle seating assembly **28**, but rather a range of positions that fall within an occupy-able subset of possible positions of the vehicle seating assembly **28**. For example, the longitudinal position of the vehicle seating assembly **28** relative to the lower track **72** and/or the angular relationship between the seatback **40** and the seat **36** may be adjusted according to the user's comfort preferences while remaining in the use state. A design position of the vehicle seating assembly **28** may be a specific longitudinal position of the vehicle seating assembly **28** relative to the lower track **72** and/or a specific angular relationship between the seatback **40** and the seat **36**. For example, the specific longitudinal position of the vehicle seating assembly **28** relative to the lower track **72** and the specific angular relationship between the seatback **40** and the seat **36** may represent a preferred arrangement of the vehicle seating assembly **28** for a majority of consumers, based on market research, population data, or other information. The vehicle seating assembly **28** is movable along the lower track **72** between a full rearward use position (line **96**) and a full forward use position (line **100**). A design longitudinal position (line **104**) is positioned at an intermediate location between the full rearward use position and the full forward use position. In various examples, the design longitudinal position may be positioned closer to the full forward use position than the full rearward use position (i.e., not a center point therebetween). The design longitudinal position of the vehicle seating assembly **28** and a design angular relationship between the seat **36** and the seatback **40** is depicted in phantom in FIG. 2A.

(35) Referring again to FIGS. 2A-2C, a distance **108** between the design longitudinal position and the full forward use position may be about 5.0 centimeters, about 10.0 centimeters, about 15.0 centimeters, about 20.0 centimeters, about 25.0 centimeters, and/or combinations or ranges thereof. Similarly, a distance **112** between the design longitudinal position and the full rearward use position may be about 5.0 centimeters, about 10.0 centimeters, about 15.0 centimeters, about 20.0 centimeters, about 25.0 centimeters, and/or combinations or ranges thereof. A length of the occupancy region **92** (e.g., from line **96** to line **100**) may be about 10.0 centimeters, about 20.0 centimeters, about 30.0 centimeters, about 40.0 centimeters, about 50.0 centimeters, and/or combinations or ranges thereof. Actuation of the vehicle seating assembly **28** along the lower track **72** may be initiated by actuation of one or more handles. For example, a first handle **116** may be actuated to initiate moving the vehicle seating assembly **28** from the use state toward the first non-use state. A second handle **120** may be actuated to initiate moving the vehicle seating assembly **28** from the use state toward the second non-use state. A third handle **124** may be actuated to initiate moving the vehicle seating assembly **28** along the lower track **72**.

(36) Referring further to FIGS. 2A-2C, in some examples, the third handle **124** may be the only one of the handles that enables movement of the vehicle seating assembly **28** longitudinally along the lower track **72**. Alternatively, actuation of the first handle **116** and/or the second handle **120** may simultaneously result in actuation of components of the vehicle seating assembly **28** that are associated with the third handle **124**. Said another way, the first handle **116**, the second handle **120**, and the third handle **124** may each be capable of enabling actuation of the vehicle seating assembly **28** along the lower track **72**. Accordingly, in such examples, the first handle **116** and the second handle **120** can each perform at least two functions when actuated. These at least two functions include moving the vehicle seating assembly **28** toward the first non-use state (actuation of the first handle **116**) or the second non-use state (actuation of the second handle **120**), as well as enabling actuation of the vehicle seating assembly **28** along the lower track **72**. In contrast, the third handle

124 may be mono-functional in that actuation of the third handle **124** may only enable actuation of the vehicle seating assembly **28** along the lower track **72** (e.g., by disengaging the upper track **76** from the lower track **72**). In various examples, the first handle **116** may be positioned within an upper region of a rear side of the seatback **40**, the second handle **120** may be positioned within a lower region of the rear side of the seatback **40** or a rearward region of the seat **36**, and the third handle **124** may be positioned in a forward region of the seat **36**. However, the present disclosure is not limited to such positioning of the first handle **116**, the second handle **120**, and/or the third handle **124**. An actuation force **128** is shown applied to each of the first, second, and third handles **116**, **120**, and **124**. Movement arrows **132** are also shown in FIGS. 2B and 2C to denote one example of how application of the given actuation force **128** may induce movement of at least one component of the vehicle seating assembly **28**.

(37) Referring still further to FIGS. 2A-2C, the track assembly **68** may enable movement of the vehicle seating assembly **28** along the lower track **72** beyond the occupancy region **92** and into one or more non-occupancy regions **136**. The non-occupancy region **136** may be positioned forward of the occupancy region **92**, as depicted in FIGS. 2B and 2C, and/or rearward of the occupancy region **92**. When the vehicle seating assembly **28** is positioned within the non-occupancy region **136**, the cam assembly **44** can restrict movement of one or more components of the vehicle seating assembly **28** to prevent a user from occupying the vehicle seating assembly **28**. For example, when the vehicle seating assembly **28** is positioned within the non-occupancy region **136**, the cam assembly **44** may aid in retaining the vehicle seating assembly **28** in the first non-use state or the second non-use state. The movement of the vehicle seating assembly(ies) **28** discussed herein may apply to outboard vehicle seating assemblies, such as the left and right vehicle seating assemblies **28** depicted in FIG. 1. However, the movement attributed to the third handle **124** (actuation along the lower track **72**) may also be applicable to vehicle seating assemblies that are in a central position (i.e., between the outboard vehicle seating assemblies).

(38) Referring now to FIGS. 3 and 4, in some examples, the mounting bracket **32** may be integrally formed or unitarily formed with the anchor **88**. In various examples, the seatback **40** may be coupled to the mounting bracket **32** and the seat **36** may be coupled to the anchor **88**. In some examples, the anchor **88** may be a portion of the mounting bracket **32** that directly couples to the upper track **76**. The vehicle seating assembly **28** is provided with an actuation cam **140**. The actuation cam **140** is configured to engage with a stationary structure below the vehicle seating assembly **28**. The stationary structure may be a portion of the floor **64** of the passenger compartment **24**. Alternatively, the stationary structure may be at least a portion of the track assembly **68** to which the vehicle seating assembly **28** is coupled. For example, the stationary structure may be a portion of the lower track **72**. A position of the vehicle seating assembly **28** relative to the stationary structure may dictate a position of the actuation cam **140**, as will be discussed in further detail herein. The actuation cam **140** includes an attachment member **144** that is configured to couple with a cable (not shown). The cable that is coupled to the attachment member **144** is communicatively coupled to the lockout cam **52** such that the position of the actuation cam **140** influences whether the lockout cam **52** is in the engaged position or a disengaged position. The actuation cam **140** is directly coupled to an actuation arm **148** such that the actuation cam **140** and the actuation arm **148** move in a coordinated fashion. The actuation arm **148** is coupled to a biasing member **152** that provides a biasing force thereto. The biasing force provided to the actuation arm **148** by way of the biasing member **152** biases the actuation arm **148** to a raised position. The raised position of the actuation arm **148** corresponds with a down position of the actuation cam **140**.

(39) Referring to FIGS. 5 and 6, the actuation cam **140** is configured to disengage with a top surface **156** of the stationary structure (e.g., the floor **64** or the lower track **72**) when the vehicle seating assembly **28** is in at least one of the first non-use state and the second non-use state. In some examples, the actuation cam **140** may fully disengage with the top surface **156** when the

vehicle seating assembly **28** is in the first non-use state and/or the second non-use state such that no physical contact remains between the actuation cam **140** and any portion of the stationary structure (e.g., see FIG. 5). In alternative examples, the actuation cam **140** may maintain contact with a portion of the stationary structure that is not the top surface **156** when the vehicle seating assembly **28** is in the first non-use state and/or the second non-use state. In some examples, the actuation cam **140** may maintain contact with a top surface of a portion of the stationary structure that is located outside of the occupancy region **92** when the vehicle seating assembly **28** is in the first non-use state and/or the second non-use state.

(40) Referring again to FIGS. 5 and 6, the track assembly **68** is provided with a track release button **160** and an interlock release button **164**. Actuation of the track release button **160** can enable movement of the upper track **76**, and ultimately the vehicle seating assembly **28**, relative to the lower track **72** within the occupancy region **92**. For example, actuation of the track release button **160** may be initiated by actuation of the third handle **124**. The first non-use state results in actuation of the track release button **160** and the interlock release button **164** of the track assembly **68** to which the vehicle seating assembly **28** is configured to be mounted. More specifically, after initiating entrance into the first non-use state (e.g., by actuation of the first handle **116**), the actuation arm **148** depresses the track release button **160** and an actuation lever **168** depresses the interlock release button **164**. For example, upon initiation of entrance into the first non-use state, an easy entry release cam **172** may be rotated toward the upper track **76** such that the actuation arm **148** and the actuation lever **168** are each also rotated toward the upper track **76**. The actuation arm **148** and the actuation lever **168** depress the interlock release button **164** and track release button **160**, respectively, when rotated toward the upper track **76**. The easy entry release cam **172** includes an engagement portion **176** that physically contacts the actuation arm **148** and the actuation lever **168** as the easy entry release cam **172** is rotated toward the upper track **76**. The engagement portion **176** of the easy entry release cam **172** extends from the actuation lever **168** to the actuation arm **148** such that the engagement portion **176** may enable simultaneous movement of the actuation arm **148** and the actuation lever **168**.

(41) Referring further to FIGS. 5 and 6, the second non-use state results in actuation of the interlock release button **164**. For example, the interlock release button **164** may be actuated in the second non-use state while the track release button **160** is not actuated in the second non-use state. Actuation of the track release button **160** independently from the interlock release button **164** is accomplished by rotation of the actuation lever **168** toward the upper track **76** in a manner that will be described further herein. The interlock release cam **180** may be sized and/or positioned such that the interlock release cam **180** is incapable of inducing actuation of the actuation lever **168**. Initiating entrance into the second non-use state may be accomplished by actuation of the second handle **120**.

(42) Referring now to FIGS. 7-8B, the vehicle seating assembly **28** is shown in the design position, according to one example. The seat **36** and the actuation arm **148** are shown in phantom to reveal neighboring components for discussion. In the depicted design position, the upper track **76** is retained in the given position relative to the lower track **72** by the track release button **160** and the interlock release button **164**. Additionally, the actuation arm **148**, the actuation lever **168**, the easy entry release cam **172**, and the interlock release cam **180** are each in raised positions. When the actuation arm **148**, the actuation lever **168**, the easy entry release cam **172**, and the interlock release cam **180** are each in raised positions, no direct physical contact may be present with each of these components and the upper track **76**. In the design position, the lockout cam **52** is free of direct physical contact with the protrusion **48** of the seatback **40** and free of direct physical contact with the seat **36**. The second end **60** of the lockout cam **52** defines a cable aperture **184** therein. The cable (not shown) that is configured to couple with the cable aperture **184** passes through a first conduit **188** that is positioned proximate to the lockout cam **52** and extends to couple with the attachment member **144** (see FIG. 5) of the actuation cam **140**. Accordingly, movement of the

actuation cam **140**, which is induced based upon a positioning of the vehicle seating assembly **28** relative to the occupancy region **92**, is translated to the lockout cam **52** by way of the cable extending therebetween. As depicted in FIG. 7, the lockout cam **52** is biased in the clockwise direction (e.g., with a clock spring). Therefore, when the actuation cam **140** is in the position depicted in FIG. 15A, a pulling force may be provided to the cable such that clockwise bias provided to the lockout cam **52** may be counter acted.

(43) Referring again to FIGS. 7-8B, the first end **56** of the lockout cam **52** defines a recess **200** therein such that the first end **56** is generally hook-shaped. The recess **200** of the first end **56** is configured to receive the protrusion **48**. A central region **204** of the lockout cam **52** receives a fastener that couples the lockout cam **52** to the mounting bracket **32**. A biasing member, such as a clock spring, may be positioned at the central region **204** to provide the clockwise biasing force to the lockout cam **52**. The central region **204** may be nearer to the mounting bracket **32** than the first end **56** and a portion of the second end **60**. In other words, the lockout cam **52** may be contoured along a length thereof that extends between the first end **56** and the second end **60** such that a distance between adjacent surfaces of the mounting bracket **32** and the lockout cam **52** is not constant along the length of the lockout cam **52**. The portion of the second end **60** that is spaced further from the mounting bracket **32** than the central region **204** may be referred to as a shelf **208**. The portions of the lockout cam **52** that are configured to engage with, and retain positioning of, other components of the vehicle seating assembly **28** (e.g., the first end **56** and the shelf **208**) can be positioned further from the mounting bracket **32** than the central region **204**.

(44) Referring to FIGS. 9A and 9B, a cable (not shown) extends from the third handle **124** to a coupling member **212** of a leg **216** of the actuation lever **168**. Actuation of the third handle **124** results in a pulling force being applied to the cable that extends between the third handle **124** and the coupling member **212**. The pulling force applied to the cable that extends between the third handle **124** and the coupling member **212** results in movement of the leg **216** toward a conduit **220** that the cable passes through, as indicated by arrow **224**. The pulling force applied to the leg **216** results in rotation of the actuation lever **168**, as indicated by arrow **228**. Rotation of the actuation lever **168** toward the upper track **76** results in actuation of the track release button **160** to a lowered, or depressed, position, as indicated by arrow **232**. Such actuation of the track release button **160** enables the upper track **76**, and ultimately the vehicle seating assembly **28**, to be adjusted in its position along the lower track **72** within the occupancy region **92**. Actuation of the third handle **124** can initiate the actuation lever **168** to disengage the track release button **160** without movement of the easy entry release cam **172**. The progression of the components as described in FIGS. 9A and 9B may be referred to as an occupant comfort adjust functionality.

(45) Referring now to FIGS. 10-12, with preceding figures, the seat **36** and the actuation arm **148** are shown in phantom to enable depiction of adjacent structures and components. The second non-use state is depicted. In the second non-use state, the engaged position of the lockout cam **52** is defined as physical contact between the second end **60** of the lockout cam **52** and the seat **36**. For example, the shelf **208** of the second end **60** of the lockout cam **52** can directly engage with a rear portion **236** of the seat **36**. In such an example, the direct physical contact between the shelf **208** and the rear portion **236** may represent the engaged position of the lockout cam **52** for the second non-use state. The engagement between the lockout cam **52** and the seat **36** retains the seat **36** in a position that prevents occupancy of the vehicle seating assembly **28** for the second non-use state. The second non-use state may be referred to as a seat-up cargo stow state. The seat **36** may be biased toward the position depicted in FIG. 10 such that actuation of the second handle **120** releases one or more components of the vehicle seating assembly **28** and permits a biasing force to rotate the seat **36** toward the seatback **40**. In the second non-use state, rotation of the seat **36** toward the seatback **40**, as indicated by arrow **240**, results in a pulling force, indicated at arrow **244**, being applied to a cable **246** that extends from a seat spool **248** to a coupling member **196** of the interlock release cam **180**. The cable **246** passes through a second conduit **252** that is positioned proximate to

the interlock release cam **180**.

(46) Referring again to FIGS. **10-12**, the pulling force applied to the cable **246** causes the coupling member **196** to move toward the second conduit **252**, as indicated by arrow **256**. The movement of the coupling member **196** toward the second conduit **252** induces rotation of the interlock release cam **180** toward the upper track **76**, as indicated by arrow **260**. The rotation of the interlock release cam **180** toward the upper track **76** that is indicated by arrow **260** results in physical contact between the interlock release cam **180** and the actuation arm **148**. Accordingly, rotation of the interlock release cam **180** toward the upper track **76** induces rotation of the actuation arm **148** toward the upper track **76**, as indicated by arrow **264**. The rotation of the actuation arm **148** toward the upper track **76** results in actuation of the interlock release button **164** to the lowered, or disengaged, position, as indicated by arrow **268**. Biasing members, such as coil springs **272**, may be provided that are configured to bias the actuation arm **148** and the interlock release cam **180** to raised positions (e.g., see FIG. **5**).

(47) Referring further to FIGS. **10-12**, as the vehicle seating assembly **28** is moved forward, as indicated by arrow **276**, the actuation cam **140** disengages with the top surface **156** of the stationary structure. The actuation cam **140** may disengage with the top surface **156** of the stationary structure when the vehicle seating assembly **28** is moved outside of the occupancy region **92** and into the non-occupancy region **136**. In various examples, the stationary structure may be the lower track **72** of the track assembly **68**. Disengaging the actuation cam **140** with the top surface **156** of the stationary structure causes a slackening of the cable that extends between the actuation cam **140** and the lockout cam **52**. Such slackening allows the biasing force applied to the lockout cam **52** to act upon the cable extending between the lockout cam **52** and the actuation cam **140** such that the lockout cam **52** may enter the engaged position. Accordingly, the actuation cam **140** is caused to rotate counter-clockwise and the lockout cam **52** rotates clockwise as the cable moves through the first conduit **188** and a third conduit **280**. The clockwise rotation of the lockout cam **52**, as indicated by arrow **290**, results in the physical contact between the shelf **208** and the rear portion **236** such that the seat **36** is maintained in the second non-use state so long as the vehicle seating assembly **28** is in the non-occupancy region **136**.

(48) Referring to FIGS. **12-14B**, the first non-use state is depicted. In the first non-use state, the engaged position of the lockout cam **52** is defined as physical contact between the first end **56** of the lockout cam **52** and the protrusion **48** of the seatback **40**. More specifically, in the first non-use state, the protrusion **48** is received within the recess **200** defined by the first end **56** of the lockout cam **52**. Actuation of the first handle **116** may permit the seatback **40** to rotate toward the seat **36**, as indicated by arrow **284**. In some examples, without the rotation of the seatback **40** toward the seat **36** indicated at arrow **284**, the protrusion **48** may not be able to be received within the recess **200**. The rotation of the seatback **40** toward the seat **36** can provide a pulling force to a cable **288** that extends between a seatback spool **292** and an attachment member **296** of the easy entry release cam **172**. The cable **288** passes through a fourth conduit **300** that is positioned proximate to the easy entry release cam **172**.

(49) Referring again to FIGS. **12-14B**, the pulling force applied to the cable **288** causes the cable **288** to move upwardly through the fourth conduit **300**, as indicated by arrow **304**. Such movement of the cable **288** causes the attachment member **296** of the easy entry release cam **172** to move toward the fourth conduit **300**, as indicated by arrow **308**. The movement of the attachment member **296** toward the fourth conduit **300** causes the easy entry release cam **172** to rotate toward the upper track **76**. As the easy entry release cam **172** is rotated toward the upper track **76**, the engagement portion **176** physically contacts the actuation arm **148** and the actuation lever **168** such that the actuation arm **148** and the actuation lever **168** are each also rotated toward the upper track **76**. Rotation of the actuation arm **148** and the actuation lever **168** toward the upper track **76** results in actuation, or depression, of the track release button **160** and the interlock release button **164**, as indicated by arrows **232** and **268**. Actuation and disengagement of the track release button **160** and

the interlock release button **164** can permit the upper track **76**, and ultimately the vehicle seating assembly **28**, to slide along the lower track **72** to the non-occupancy region **136**. For example, actuation and disengagement of the track release button **160** and the interlock release button **164** can permit the upper track **76**, and ultimately the vehicle seating assembly **28**, to slide forward as indicated by the arrow **276** such that access may be granted to a row of vehicle seating assemblies that is positioned rearward of the vehicle seating assembly **28** (e.g., a second row or a third row).

(50) Referring further to FIGS. **12-14B**, in the first non-use state, the engaged position of the lockout cam **52** is defined as physical contact between the first end **56** of the lockout cam **52** and the protrusion **48** of the seatback **40**. The first non-use state places the seatback **40** in a position that prevents a user from occupying the vehicle seating assembly **28**. As with the process described above for the second non-use state, as the actuation cam **140** disengages with the top surface **156** of the stationary structure (e.g., a portion of the floor **64** or a portion of the lower track **72**), the biasing force applied to the lockout cam **52** initiates rotation of the lockout cam **52** toward the engaged position. More specifically, in the depicted example, the biasing force of the lockout cam **52** is countered by the physical interference between the actuation cam **140** and the stationary structure (see FIG. **15A**). Accordingly, once the physical interference between the actuation cam **140** and the stationary structure is no longer present, the biasing force applied to the lockout cam **52** rotates the recess **200** toward the protrusion **48**, as indicated at arrow **312**, and ultimately provides a pulling force to the cable that extends between the cable aperture **184** and the attachment member **144**.

(51) Referring now to FIGS. **15A** and **15B**, when the lockout cam **52** is in the engaged position, whether for the first non-use state or the second non-use state, re-engagement between the actuation cam **140** and the stationary structure (e.g., the top surface **156** of the lower track **72**) initiates transitioning the lockout cam **52** to the disengaged position depicted in FIG. **15B**. Accordingly, as the vehicle seating assembly **28** and the upper track **76** are actuated rearwardly along the lower track **72**, as indicated by arrow **316**, the actuation cam **140** contacts a front surface **320** of a portion of the lower track **72**. The contact between the front surface **320** and the actuation cam **140** results in clockwise rotation of the actuation cam **140**, as indicated by arrow **324**. Therefore, when the vehicle seating assembly **28** moves from the non-occupancy region **136** to the occupancy region **92**, the actuation cam **140** is rotated clockwise and acts against the biasing force that encourages the lockout cam **52** to the engaged position.

(52) Referring again to FIGS. **15A** and **15B**, engagement between the actuation cam **140** and the top surface **156** of the stationary structure (e.g., the lower track **72**) places the actuation cam **140** in a position that provides a force (e.g., a constant force) to the cable that counter-acts the biasing of the lockout cam **52** to the engaged position. The clockwise rotation of the actuation cam **140** results in a pulling force being applied to the cable that extends between the attachment member **144** and the cable aperture **184**. The pulling force applied to the cable as a result of the clockwise rotation of the actuation cam **140** results in the second end **60** of the lockout cam **52** being pulled toward the first conduit **188**. Movement of the second end **60** of the lockout cam **52** toward the first conduit **188** causes a counter-clockwise rotation of the lockout cam **52**, as indicated by arrow **328**, which results in a disengagement between the recess **200** and the protrusion **48**. With the lockout cam **52** exiting the engaged position, the vehicle seating assembly **28** is capable of returning to an arrangement of the seat **36** and the seatback **40** that is capable of receiving an occupant.

(53) Modifications of the disclosure will occur to those skilled in the art and to those who make or use the concepts disclosed herein. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims as interpreted according to the principles of patent law, including the doctrine of equivalents.

(54) It will be understood by one having ordinary skill in the art that construction of the described concepts, and other components, is not limited to any specific material. Other exemplary

embodiments of the concepts disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.

(55) For purposes of this disclosure, the term “coupled” (in all of its forms: couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature, or may be removable or releasable in nature, unless otherwise stated.

(56) It is also important to note that the construction and arrangement of the elements of the disclosure, as shown in the exemplary embodiments, is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts, or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, and the nature or numeral of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.

(57) It will be understood that any described processes, or steps within described processes, may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

(58) It is also to be understood that variations and modifications can be made on the aforementioned structures and methods without departing from the concepts of the present disclosure, and further, it is to be understood that such concepts are intended to be covered by the following claims, unless these claims, by their language, expressly state otherwise.

Claims

1. A vehicle seating assembly, comprising: a mounting bracket; a seat that is rotatably coupled to the mounting bracket; a seatback that is rotatably coupled to the mounting bracket, wherein the seatback comprises a protrusion; a cam assembly rotatably coupled to the mounting bracket, wherein the cam assembly comprises a lockout cam having a first end and a second end, wherein the first end of the lockout cam engages with the protrusion of the seatback such that movement of the seatback relative to the mounting bracket is restricted in a first non-use state, and wherein the second end of the lockout cam engages with the seat such that movement of the seat relative to the mounting bracket is restricted in a second non-use state, wherein the lockout cam is biased toward an engaged position; an actuation cam that is configured to engage with a stationary structure below the vehicle seating assembly, wherein a position of the vehicle seating assembly relative to the stationary structure dictates a position of the actuation cam; and a cable that extends between the actuation cam and the lockout cam such that the position of the actuation cam influences

whether the lockout cam is in the engaged position or a disengaged position, wherein disengagement of the actuation cam with a top surface of the stationary structure causes a slackening of the cable that extends between the actuation cam and the lockout cam, and wherein such slackening of the cable allows the biasing of the lockout cam to place the lockout cam in the engaged position.

2. The vehicle seating assembly of claim 1, wherein the engaged position of the lockout cam is defined as physical contact between the first end of the lockout cam and the protrusion of the seatback in the first non-use state.

3. The vehicle seating assembly of claim 1, wherein the engaged position of the lockout cam is defined as physical contact between the second end of the lockout cam and the seat in the second non-use state.

4. The vehicle seating assembly of claim 1, wherein the stationary structure is at least a portion of a track assembly to which the vehicle seating assembly is configured to be mounted.

5. The vehicle seating assembly of claim 1, wherein the actuation cam is configured to disengage with a top surface of the stationary structure when the vehicle seating assembly is in at least one of the first non-use state and the second non-use state.

6. The vehicle seating assembly of claim 1, wherein engagement between the actuation cam and the top surface of the stationary structure places the actuation cam in a position that provides a force to the cable that counter-acts the biasing of the lockout cam to the engaged position.

7. The vehicle seating assembly of claim 1, wherein the first non-use state results in actuation of a track release button and an interlock release button of a track assembly to which the vehicle seating assembly is configured to be mounted.

8. The vehicle seating assembly of claim 7, wherein the second non-use state results in actuation of the interlock release button.

9. The vehicle seating assembly of claim 1, wherein the first non-use state places the seatback in a position that prevents occupancy of the vehicle seating assembly.

10. The vehicle seating assembly of claim 7, wherein the second non-use state places the seat in a position that prevents occupancy of the vehicle seating assembly.

11. A vehicle seating assembly, comprising: a mounting bracket; a seat that is rotatably coupled to the mounting bracket; a seatback that is rotatably coupled to the mounting bracket, wherein the seatback comprises a protrusion; a cam assembly rotatably coupled to the mounting bracket, wherein the cam assembly comprises a lockout cam that is biased toward an engaged position, wherein the lockout cam comprises a first end and a second end, wherein the first end of the lockout cam engages with the protrusion of the seatback such that movement of the seatback relative to the mounting bracket is restricted in a first non-use state, wherein the engaged position of the lockout cam is defined as physical contact between the first end of the lockout cam and the protrusion of the seatback in the first non-use state, wherein the second end of the lockout cam engages with the seat such that movement of the seat relative to the mounting bracket is restricted in a second non-use state, and wherein the engaged position of the lockout cam is defined as physical contact between the second end of the lockout cam and the seat in the second non-use state; an actuation cam that engages with a stationary structure below the vehicle seating assembly, wherein a position of the vehicle seating assembly relative to the stationary structure dictates a position of the actuation cam, wherein the actuation cam disengages with a top surface of the stationary structure when the vehicle seating assembly is in at least one of the first non-use state and the second non-use state, wherein engagement between the actuation cam and the top surface of the stationary structure places the actuation cam in a position that provides a force to the cable that counter-acts the biasing of the lockout cam to the engaged position; and a cable that extends between the actuation cam and the lockout cam such that the position of the actuation cam influences whether the lockout cam is in the engaged position or a disengaged position, wherein disengagement of the actuation cam with the top surface of the stationary structure causes a

slackening of the cable that extends between the actuation cam and the lockout cam, and wherein such slackening of the cable allows the biasing of the lockout cam to place the lockout cam in the engaged position.

12. The vehicle seating assembly of claim 11, wherein the stationary structure is at least a portion of a track assembly to which the vehicle seating assembly is mounted.

13. The vehicle seating assembly of claim 11, wherein the first non-use state results in actuation of a track release button and an interlock release button of a track assembly to which the vehicle seating assembly is mounted.

14. The vehicle seating assembly of claim 13, wherein the second non-use state results in actuation of the interlock release button.

15. The vehicle seating assembly of claim 11, wherein the first non-use state places the seatback in a position that prevents occupancy of the vehicle seating assembly, and wherein the second non-use state places the seat in a position that prevents occupancy of the vehicle seating assembly.
